UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 8 printed pages

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QUESTION 1

a) Write the solution of -x - 1 > -3 in interval notation and on a number line.

(3 marks)



Figure: solution set.

b) Solve the inequality |-4x - 2| ≥ 3. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Solve the equation |2 - 5x| = 9.

(4 marks)

QUESTION 2

a) Given f(x) = x + 1 and g(x) = -x - 4, find (f + g)(x), (f - g)(x) and (f g)(x). State the domain of each.

(4 marks)

b) The piecewise function is defined by
f(x) = 3x + 4, if x < -1;
f(x) = -1, if -1 ≤ x ≤ 2;
f(x) = x² + 3, if x > 2.
Sketch the graph and find f(-2), f(0) and f(3).

(5 marks)

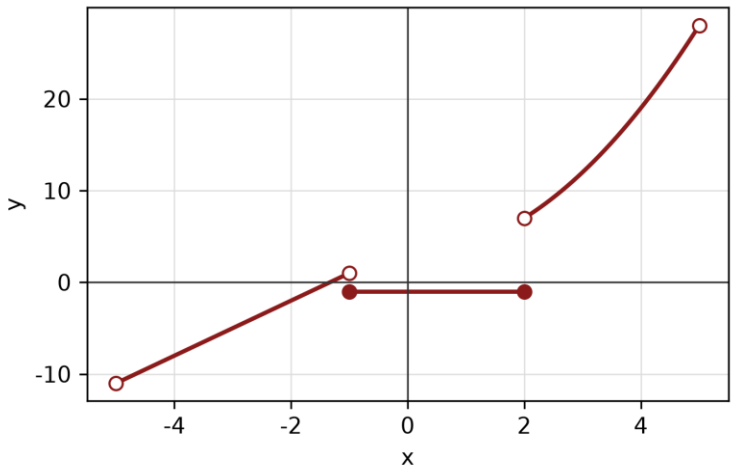


Figure: piecewise graph.

c) Write $x^2 - 8x + y^2 + 8y + 7 = 0$ in standard form. Hence state the centre and radius and sketch the circle.

(5 marks)

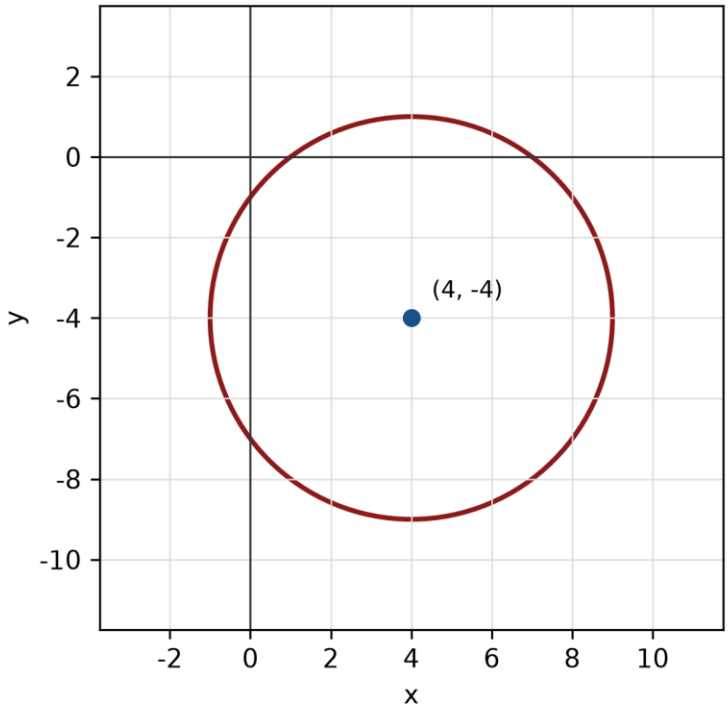


Figure: circle.

QUESTION 3

a) Given $z_1 = -5 - 6i$ and $z_2 = 4 + 6i$, find $z_1 \times z_2$ in the form $a + bi$.

(4 marks)

b) Find the modulus and argument of $z = 4 + 3i$. Give the argument in degrees correct to one decimal place.

(5 marks)

c) Factorise $f(x) = 2x^3 - 2x^2 - 24x$ completely.

(5 marks)

QUESTION 4

a) Solve the system of equations simultaneously:

$2x + 3y + z = -7$
$-x - 3y - 2z = 2$
$-2x + 2y - z = 2$

(5 marks)

b) Solve the simultaneous equations

$y = x^2 - 1$
$y = 4x - 1.$

(5 marks)

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c) Find the equation of the line passing through (0, 4) and parallel to the line -5x + 2y - 7 = 0.

(5 marks)

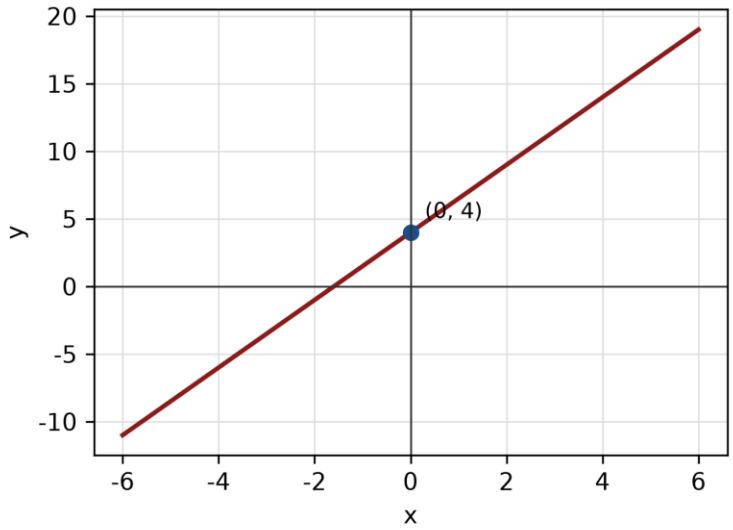


Figure: required line.

QUESTION 5

a) Given sin A = 8/17 where A is in quadrant II and cos B = 15/17 where B is in quadrant IV, find sin 2A and cos(A + B).

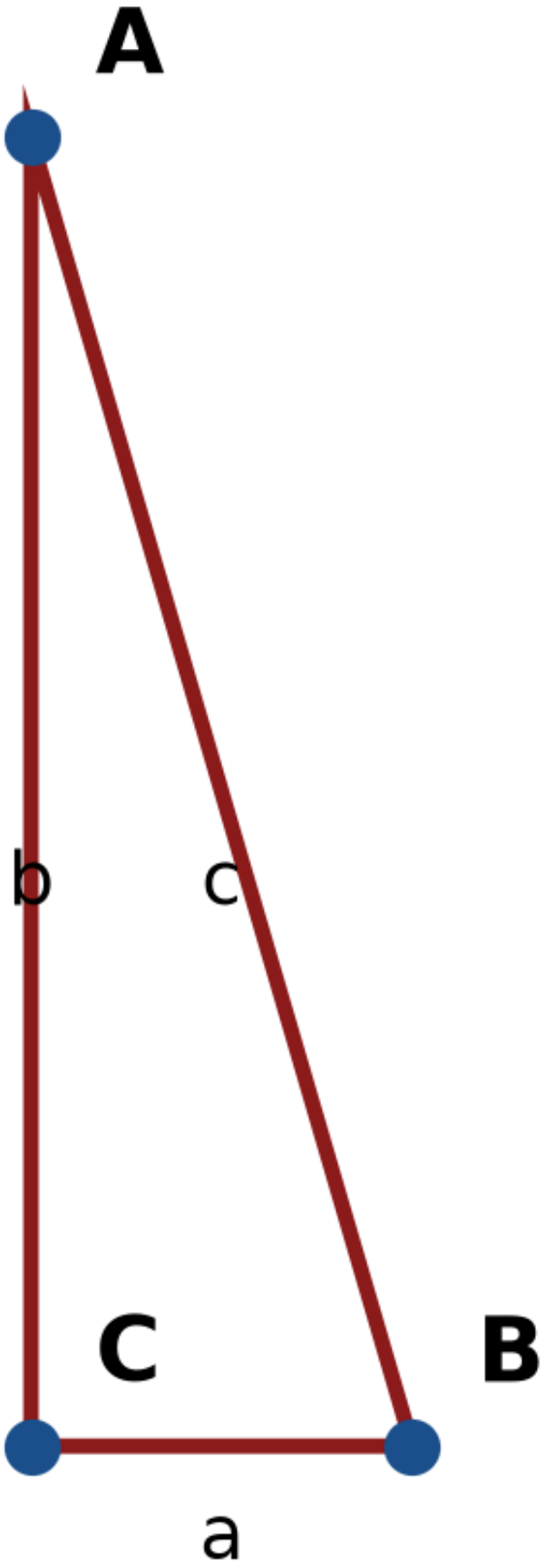
(5 marks)

b) Convert 90° to radians, and convert π/6 radians to degrees.

(3 marks)

c) Solve the triangle with sides a = 14 cm, b = 48 cm and c = 50 cm. Find all three angles.
(5 marks)

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Figure: right triangle.

END OF QUESTION PAPER

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APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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